



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · MOTOR-CUM-HAND OPERATED

CIE/777 HM

Motor-cum-hand driven insulation tester — mains motor for PI testing, hand-crank backup for field use, 500V to 10,000V.

UP TO 10000V

Motor-cum-Hand Driven Insulation Tester

Combines mains-motor drive with a collapsible hand-crank backup in one instrument — the same sustained-voltage PI and dielectric absorption capability as the CIE/777, plus a centrifugal-governed hand crank for use where mains power is unavailable.

- **Motor + hand-crank backup** — works with or without mains
- **Sustained PI & DAR testing** — motor-driven mode
- **Collapsible generator handle** — specific to the HM variant
- **Centrifugal governor** — constant voltage in hand-crank mode
- **Seasoned wooden body** — sunmica finish, detachable lid & lock
- **6 rated voltages** — 500V to 10,000V, build-to-order



- M Motor + Hand** Works with or without mains
- P PI & DAR** Sustained voltage testing
- V Up to 10000V** 6 rated voltages

Manufactured in India since 1963 — trusted by MSEB, Indian Railways, WBSETCL, etc.

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Drive	Mains motor (230V, 50Hz A.C.) + collapsible hand-crank
Body	Seasoned wooden cabinet, sunmica finish, detachable lid, handle, lock
Scale Length	100 mm (approx.)
Accuracy	As per IS: 2992-1987
High Voltage Test	Min. 2kV or (rated V + 1)kV A.C. r.m.s., 1 min
Insulation Resistance Test	Better than 50 M Ω at 500V D.C.
Response Time	3 seconds
Rated Voltages	500V / 1000V / 2500V / 5000V / 10000V
500V Range Options	0–1000 M Ω
1000V Range Options	0–2000 M Ω
2500V Range Options	0–5000 · 0–10000 · 0–20000 M Ω
5000V Range Options	0–5000 · 0–10000 · 0–20000 · 50–50000 · 100–100000 M Ω
10000V Range Options	0–10000 · 0–20000 · 50–50000 · 100–100000 M Ω
Ordering	Build-to-order — specify voltage & resistance range
Dimensions (top cover)	14 × 10 × 10 inch (approx.)
Weight (top cover)	12 kg (approx.) — vs 9 kg for plain CIE/777
Warranty	12 months against manufacturing defects

02 Typical Applications

- Field testing where mains supply is uncertain
- Sustained PI & DAR testing (motor-driven mode)
- Remote substation and railway traction commissioning
- HV transformer and cable insulation assessment

03 What's in the Box

- ✓ Mains cord, 2 metres
- ✓ 1 pair of H.V. testing leads

04 Optional Accessories

- + Rexine carrying case
- + 15-minute timer facility
- + Timer
- + Additional/special H.V. leads, 2m
- + Dual voltage & range option
- + Single voltage, dual range option

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Instruments for accuracy and
reliability since 1963.